

Unit-5

Deep Learning for Multimodal

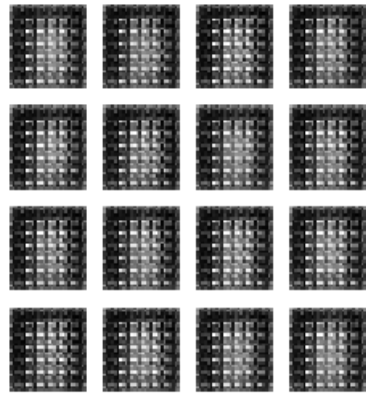
MULTIMODAL MACHINE LEARNING

MULTIMODAL AI

Bridging Language, Vision & Sound



GAN - Generative Adversarial Network



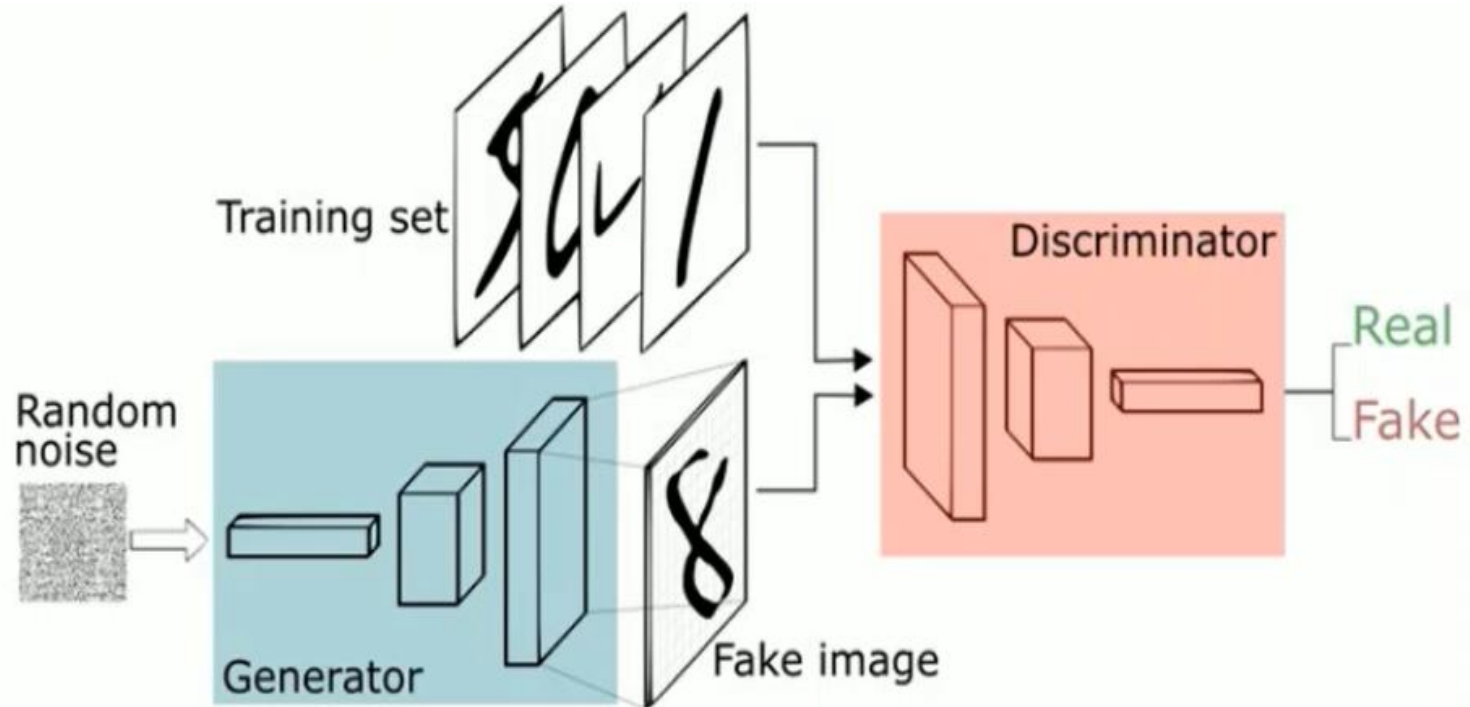
Coarse styles
($4^2 - 8^2$)

Middle styles
($16^2 - 32^2$)

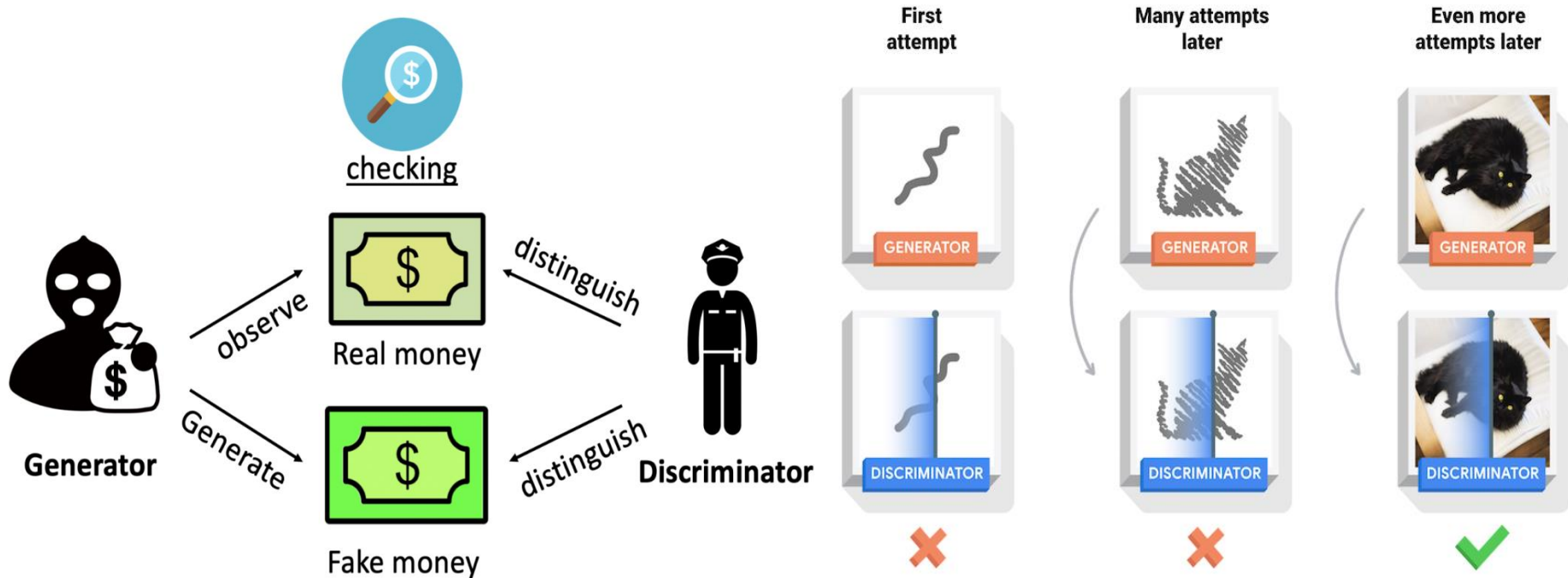
Fine styles
($64^2 - 1024^2$)



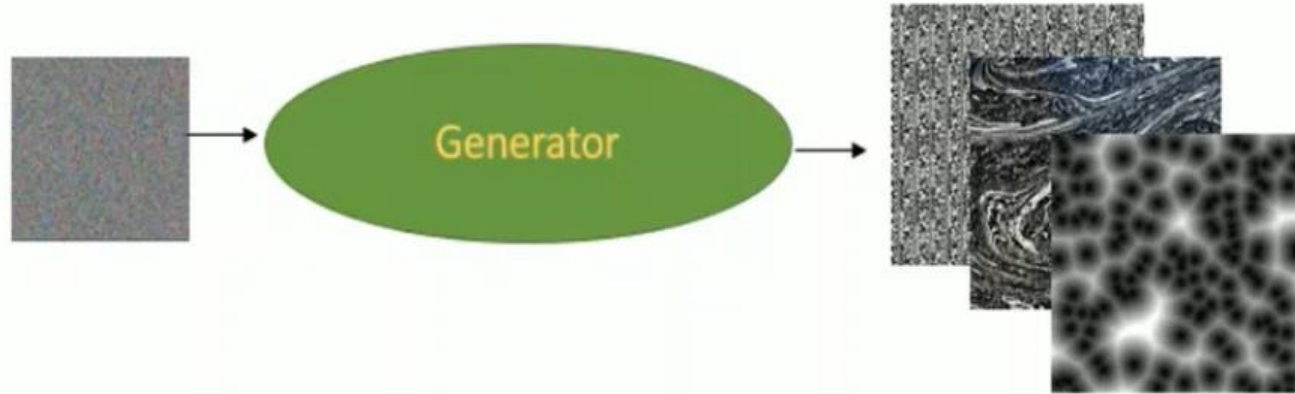
What is GAN



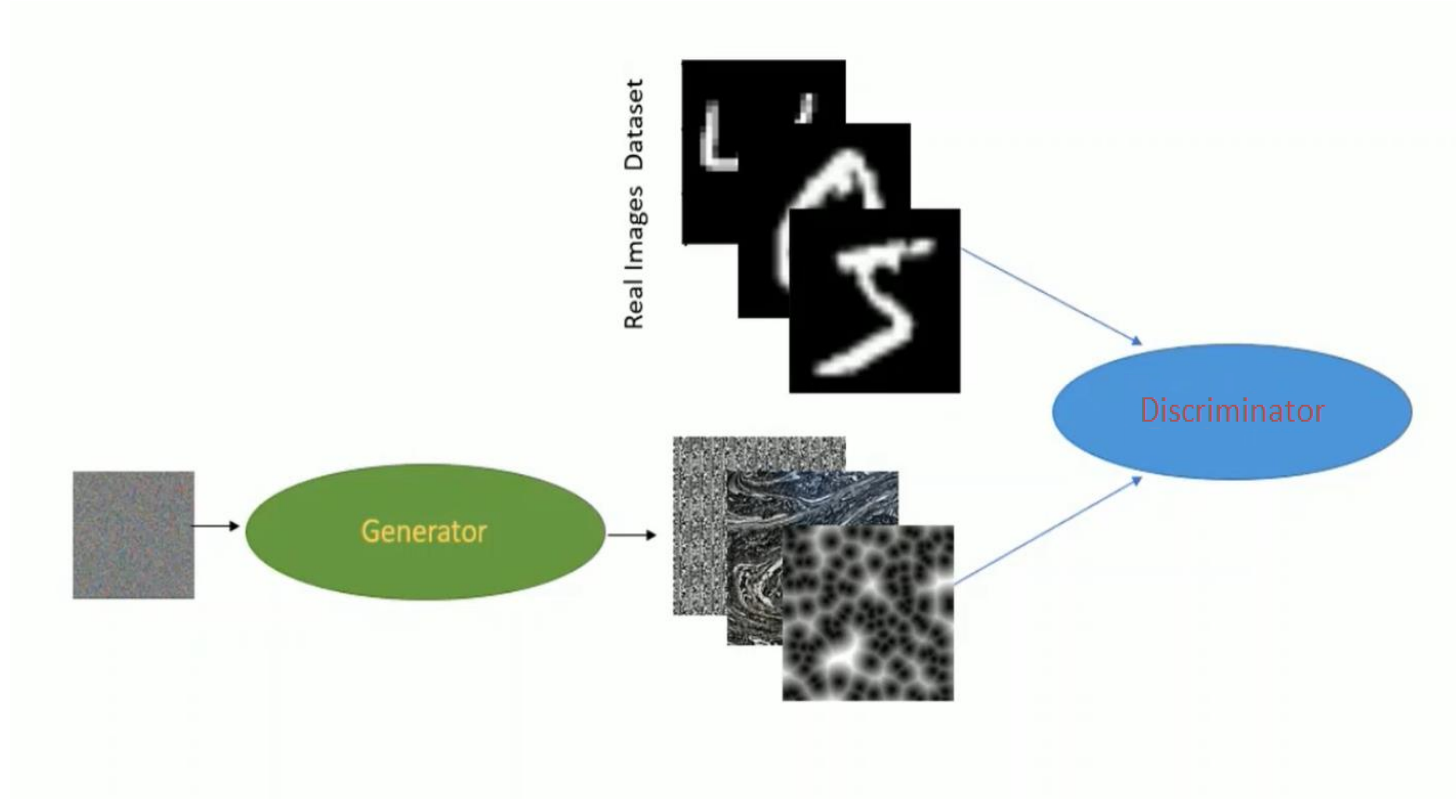
What is GAN



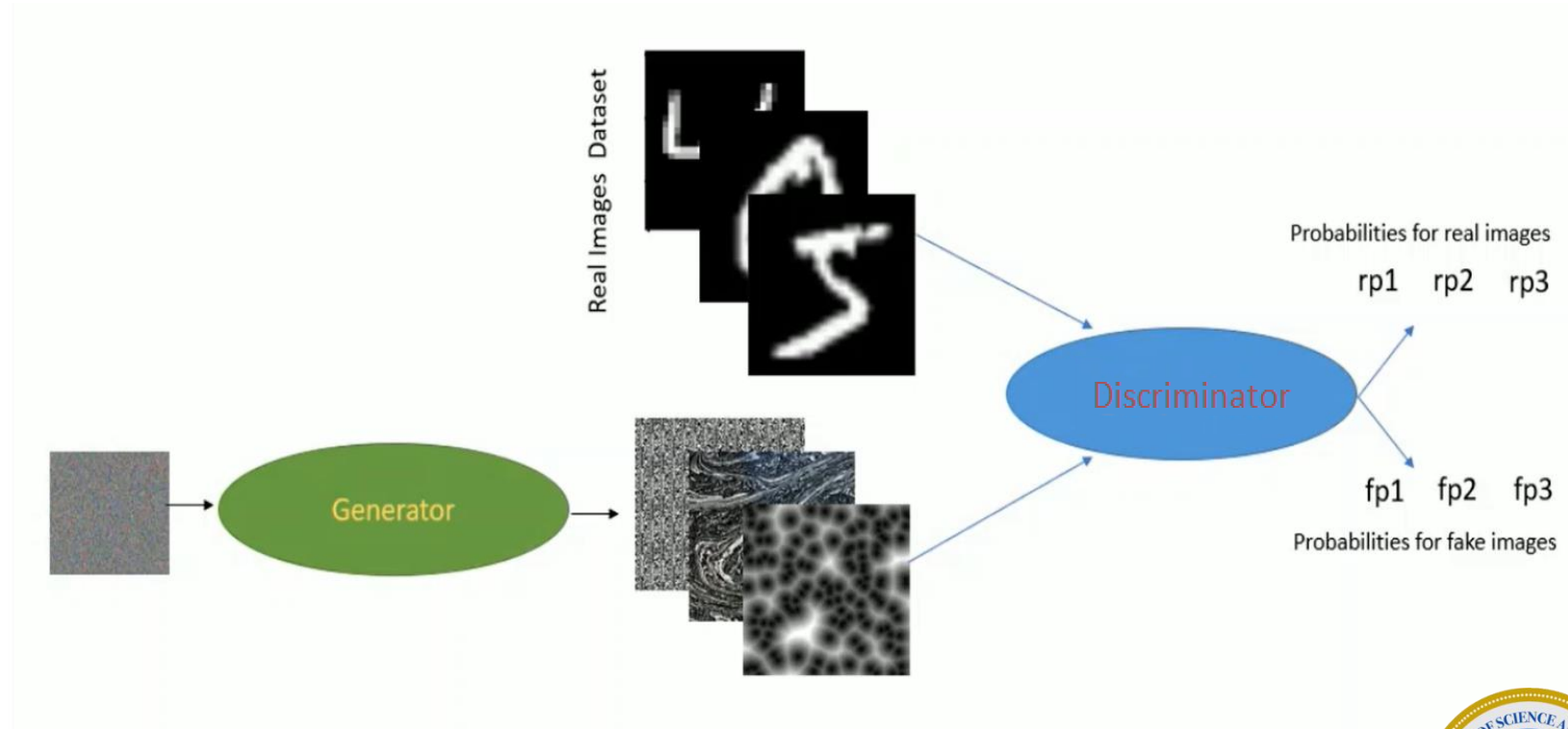
What is GAN



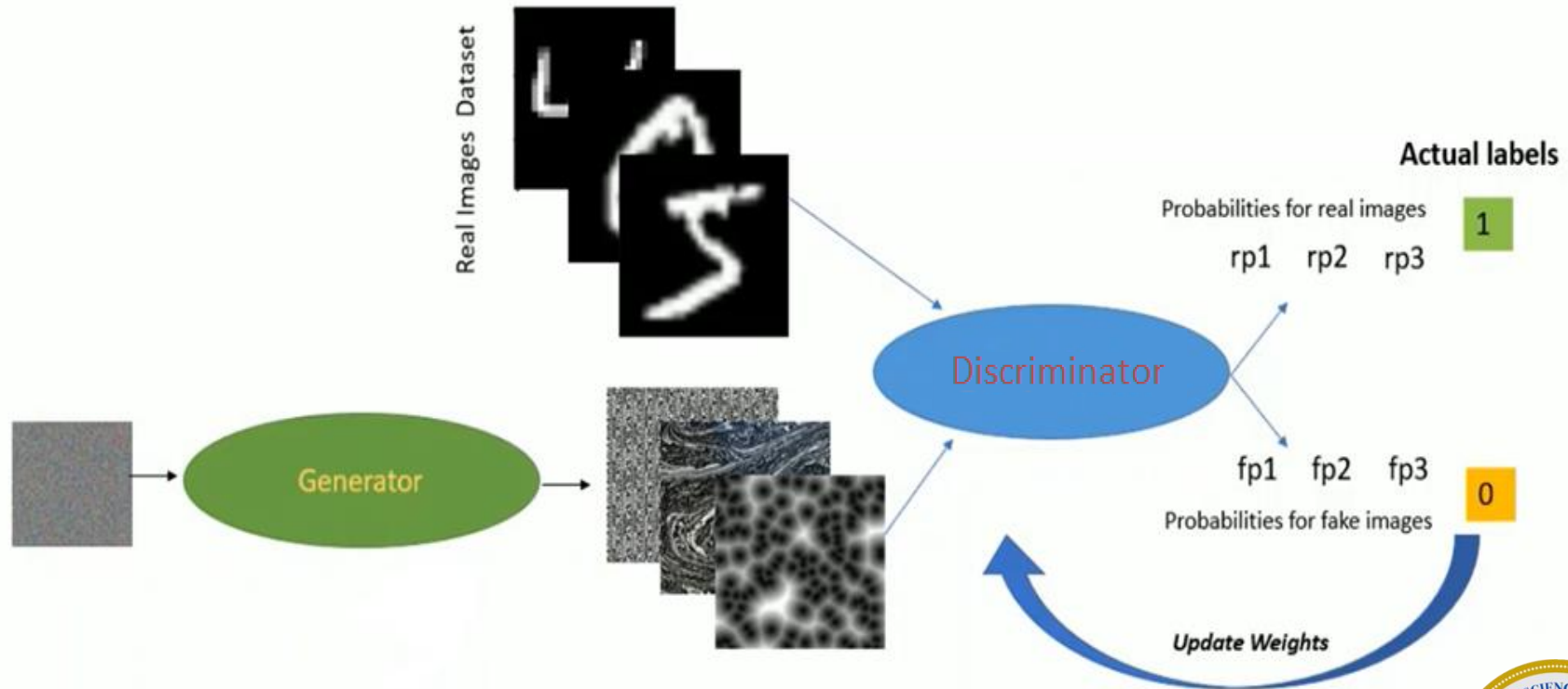
What is GAN



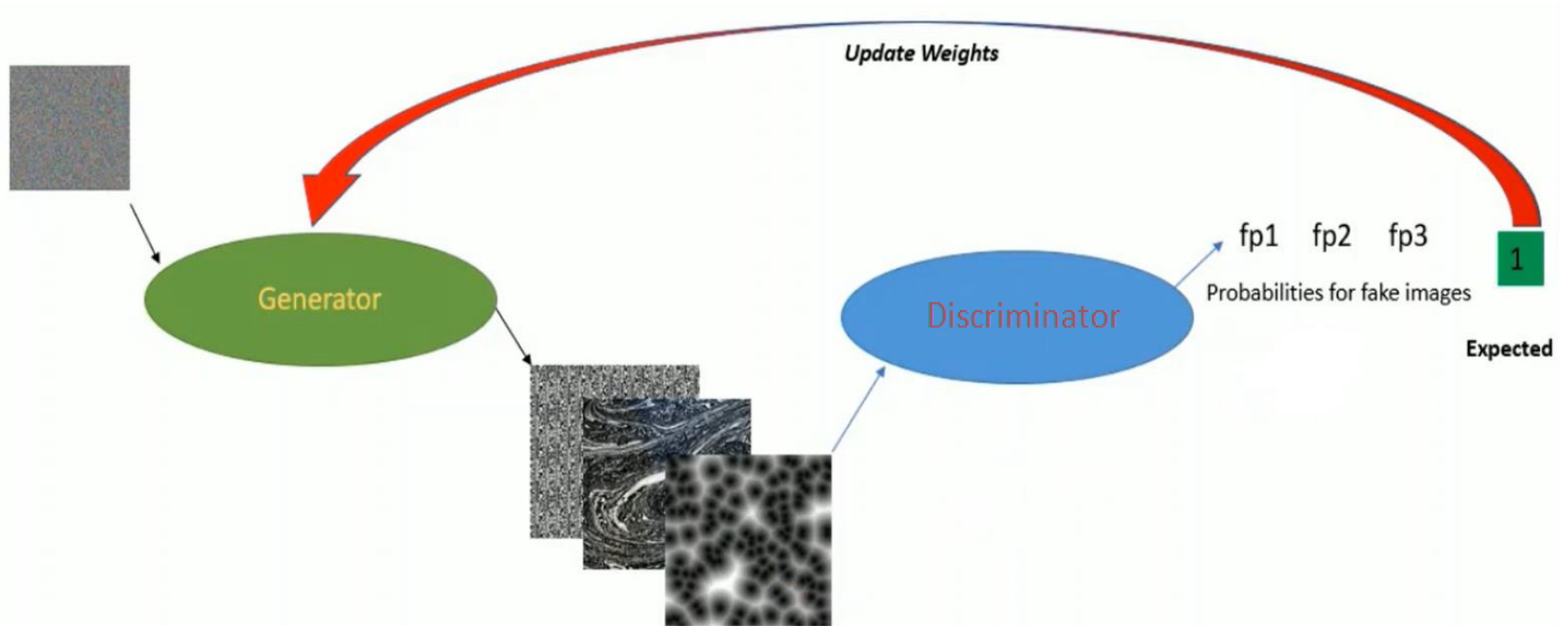
What is GAN



What is GAN



What is GAN



What is GAN

- If the D is too good, the G will never be able to fool it
- If the G is too good, it will persistently exploit weakness in the discriminator that lead to false results
- So, we use the process “Alternating Training” which is basically holding the Generator values constant when training the Discriminator and Vice Versa



Golden Rule:

learning together



Standard GAN loss function (min-max loss)

$$E_x[\log(D(x))] + E_z[\log(1 - D(G(z)))]$$

Discriminator Loss

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log (1 - D(G(z^{(i)}))) \right]$$

Generator Loss

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log (1 - D(G(z^{(i)})))$$

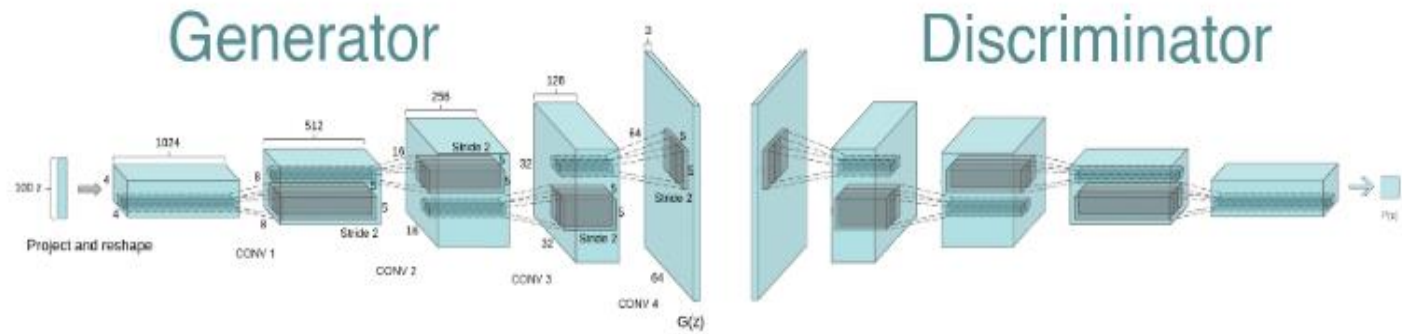


Challenges

- Vanishing Gradients
 - W loss or (Wasserstein loss) – Changing sigmoid to linear activation function
- Mode Collapse
 - [C GAN](#)
 - [Unrolled GANs](#)
- Convergence
 - [Adding noise to discriminator inputs](#)
 - [Penalizing discriminator weights](#)



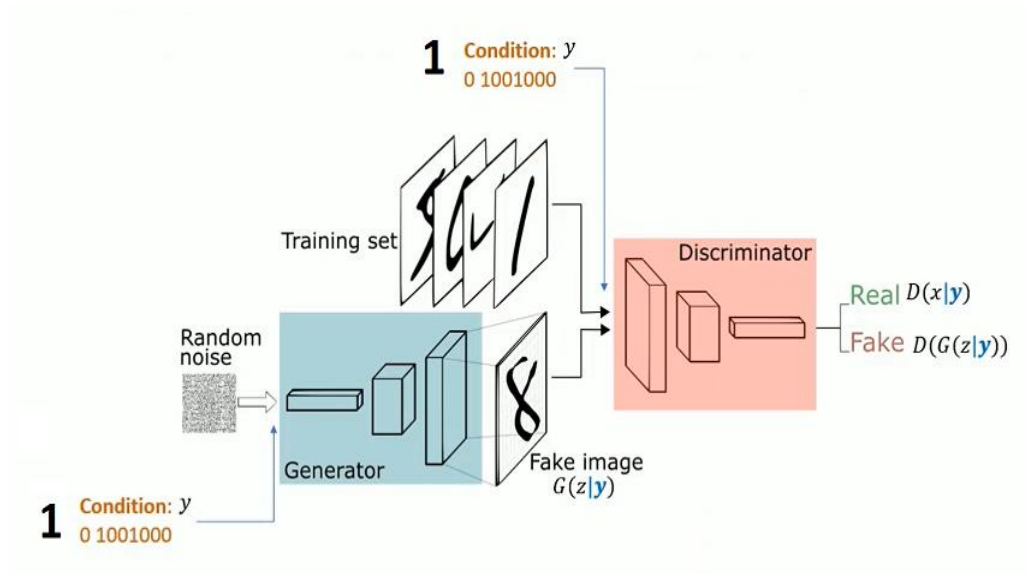
DC GAN



Conditional GAN

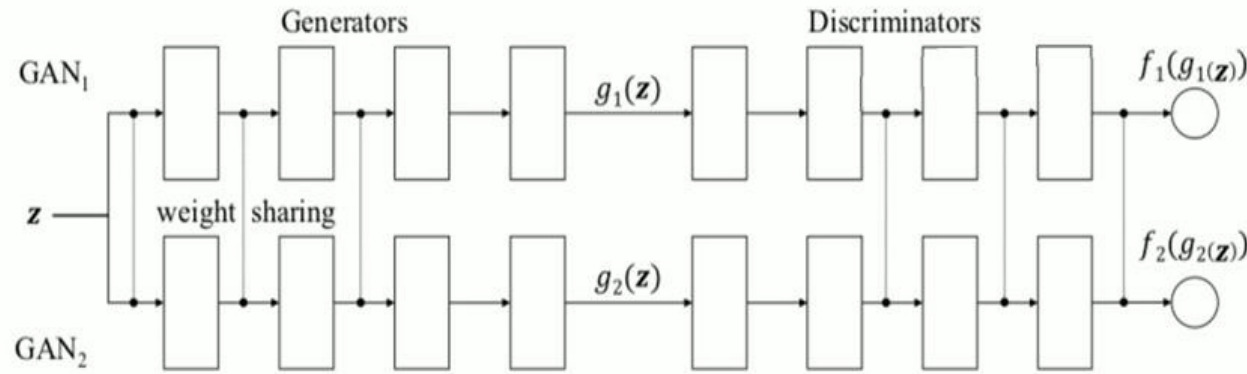
Loss

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\mathbf{z} \sim p_z(\mathbf{z})} [\log(1 - D(G(\mathbf{z}|\mathbf{y})))]$$



Coupled GAN

Just make the networks use the exact same weights for some of the layers



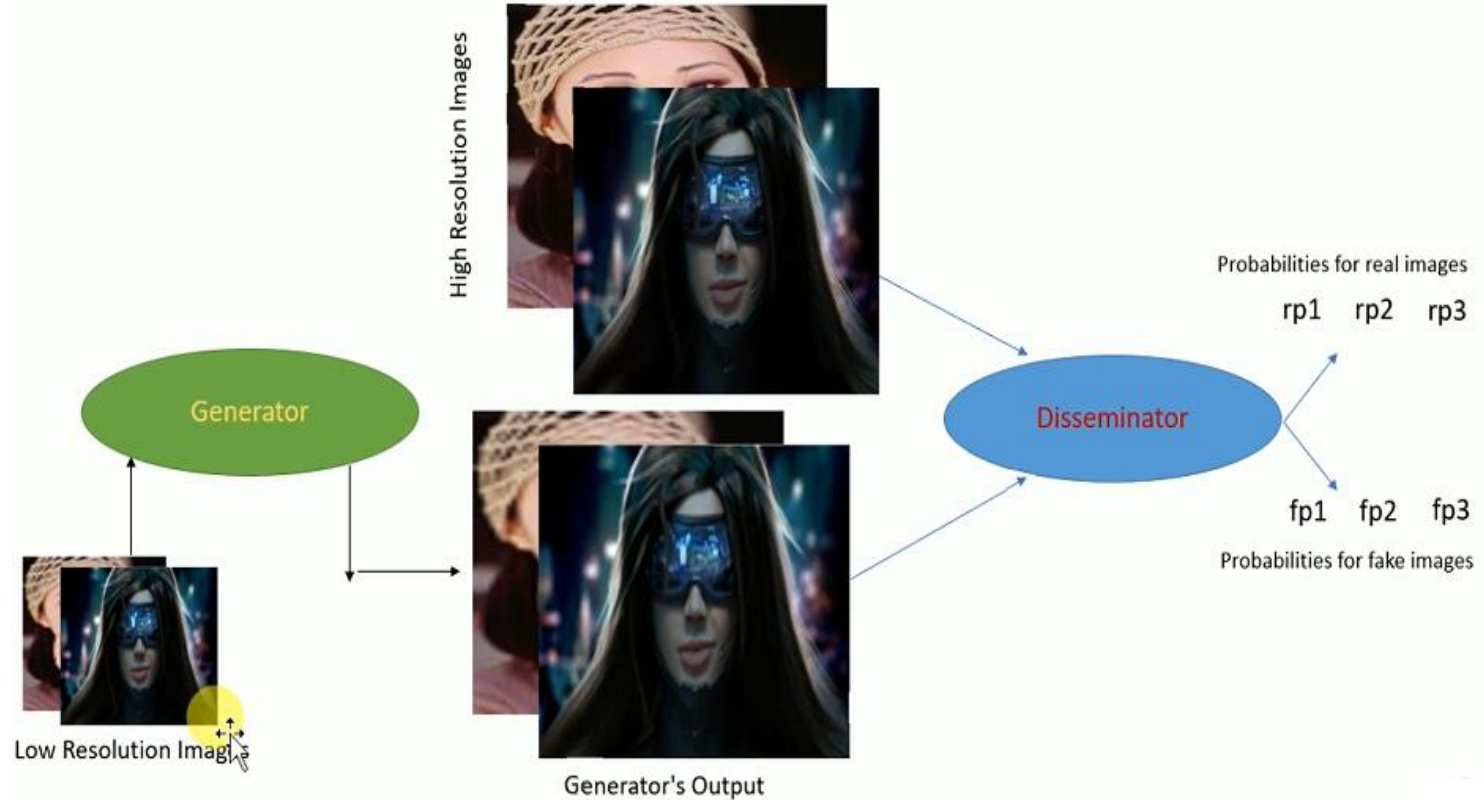
This CoGAN doesn't only increase the quality of generation but in fact you get two Images for the price of one-and-a-half. This basically save more memory, compute than two individual GANs



Super Resolution GAN



Super Resolution GAN



Super Resolution GAN

- The SRGAN uses perceptual loss function (L_{SR}) which is the weighted sum of two loss components : content loss and adversarial loss

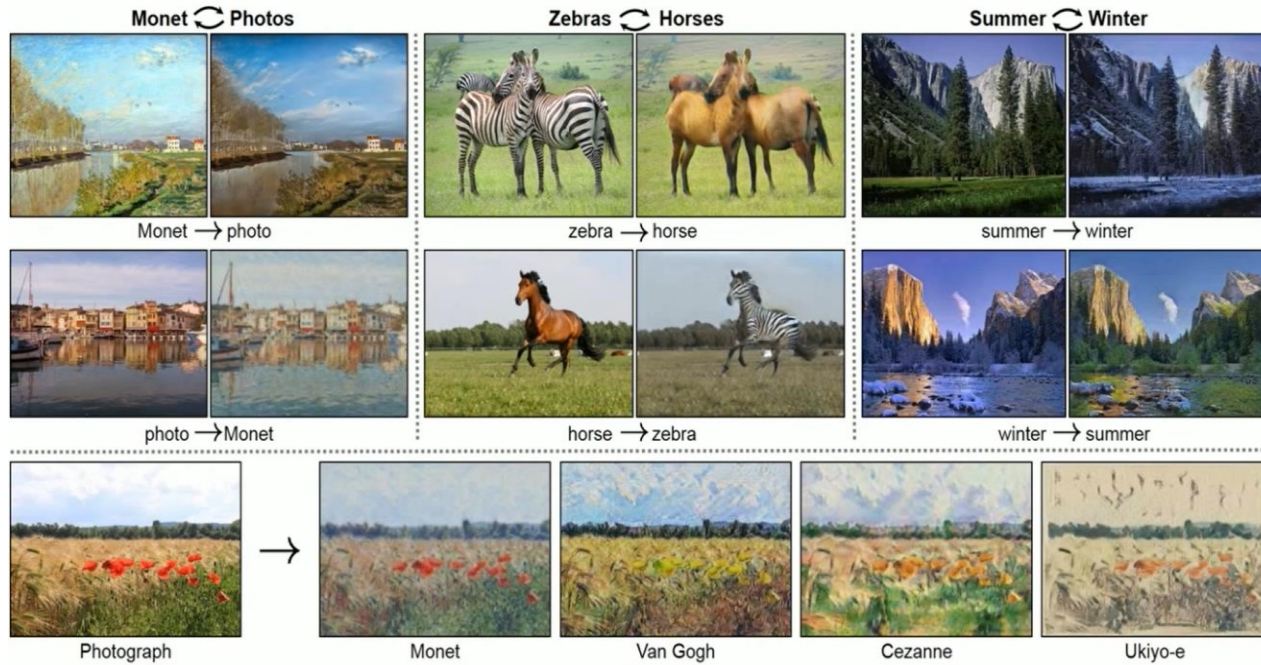
Content loss
$$l_{MSE}^{SR} = \frac{1}{r^2WH} \sum_{x=1}^{rW} \sum_{y=1}^{rH} (I_{x,y}^{HR} - G_{\theta_G}(I^{LR})_{x,y})^2$$

Adversarial Loss
$$l_{Gen}^{SR} = \sum_{n=1}^N -\log D_{\theta_D}(G_{\theta_G}(I^{LR}))$$

Overall
$$l^{SR} = \underbrace{l_X^{SR}}_{\text{content loss}} + \underbrace{10^{-3}l_{Gen}^{SR}}_{\text{adversarial loss}}$$



Cycle GAN



Cycle GAN

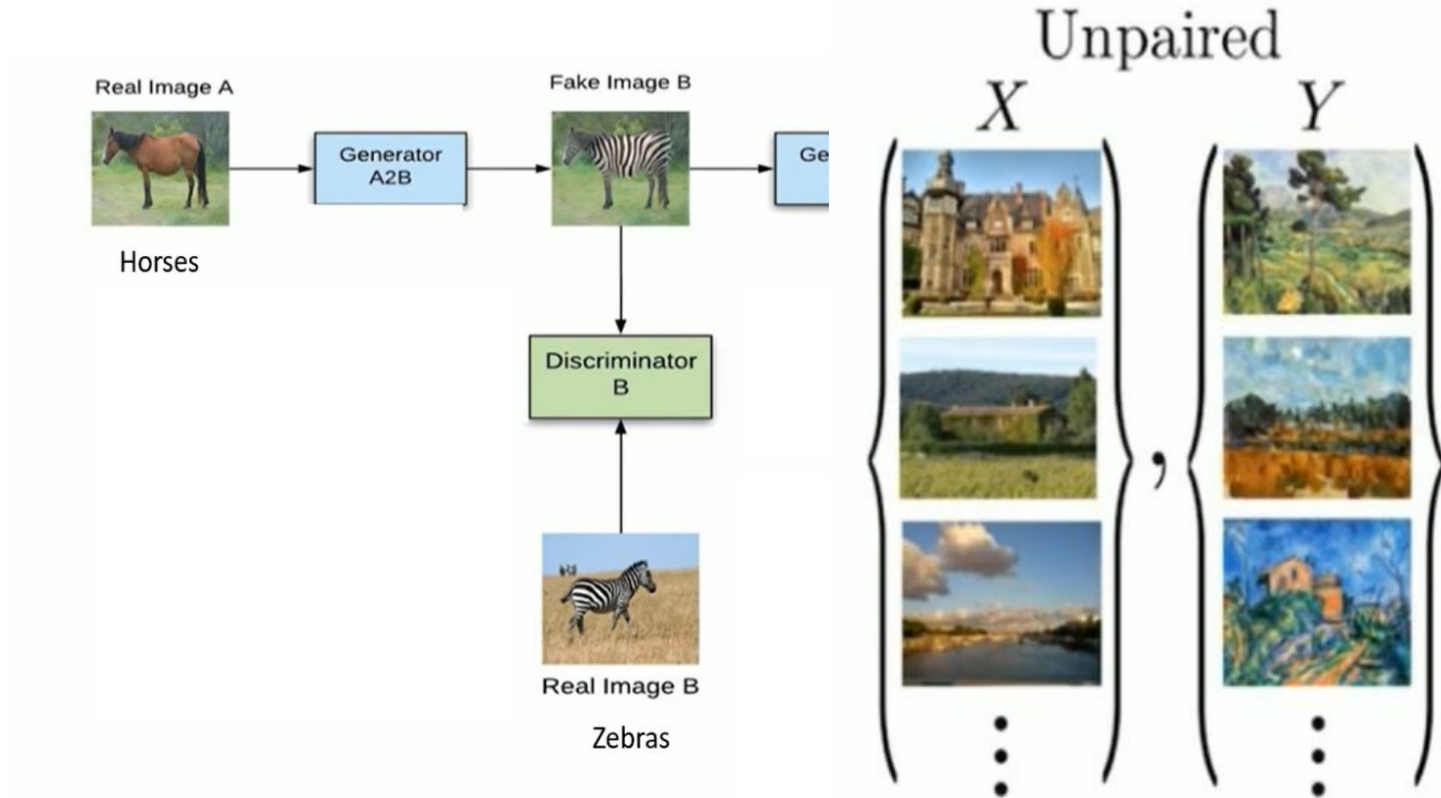
Paired Image to image Translation – Pix to Pix



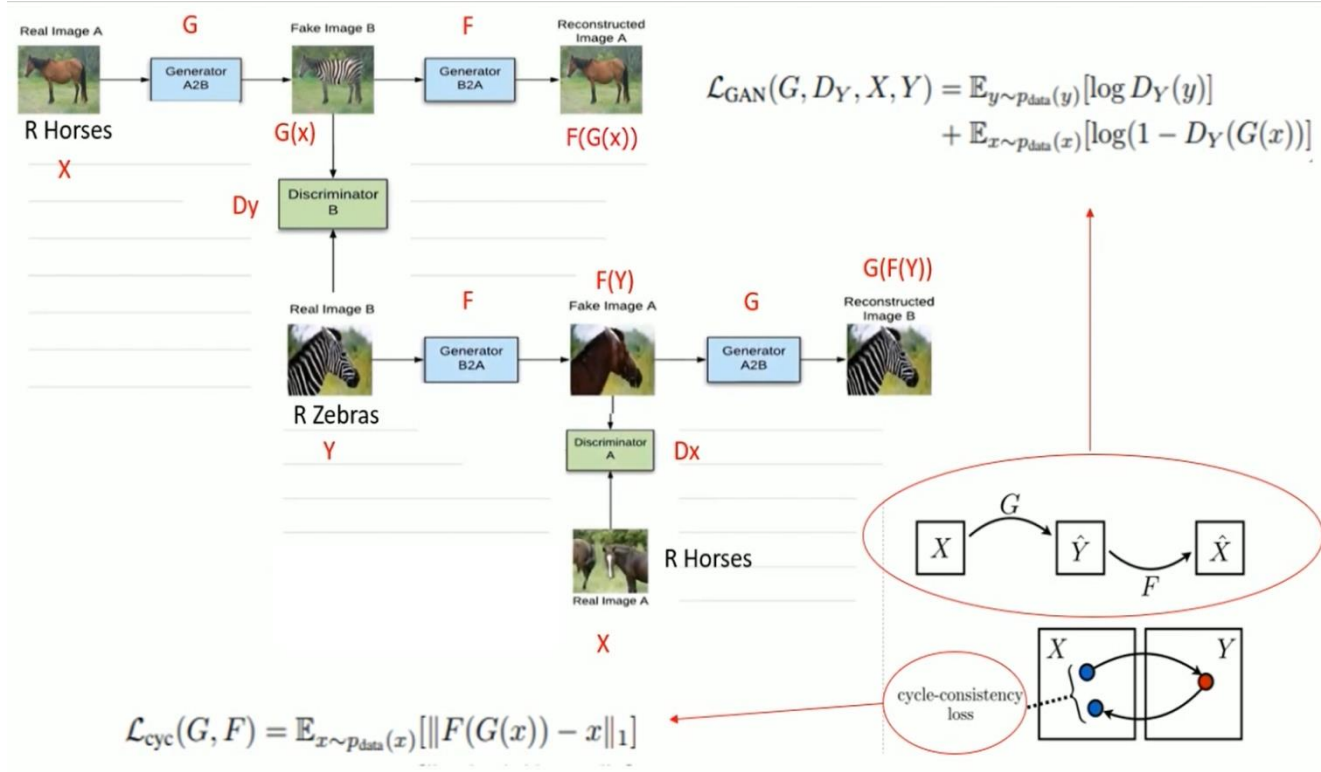
Unpaired Image to image Translation – Cycle GAN



Cycle GAN

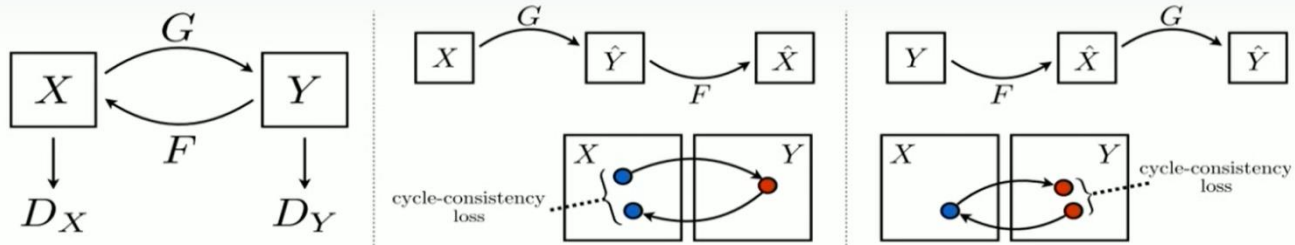


Cycle GAN



Cycle GAN

18. CycleGANs Theory



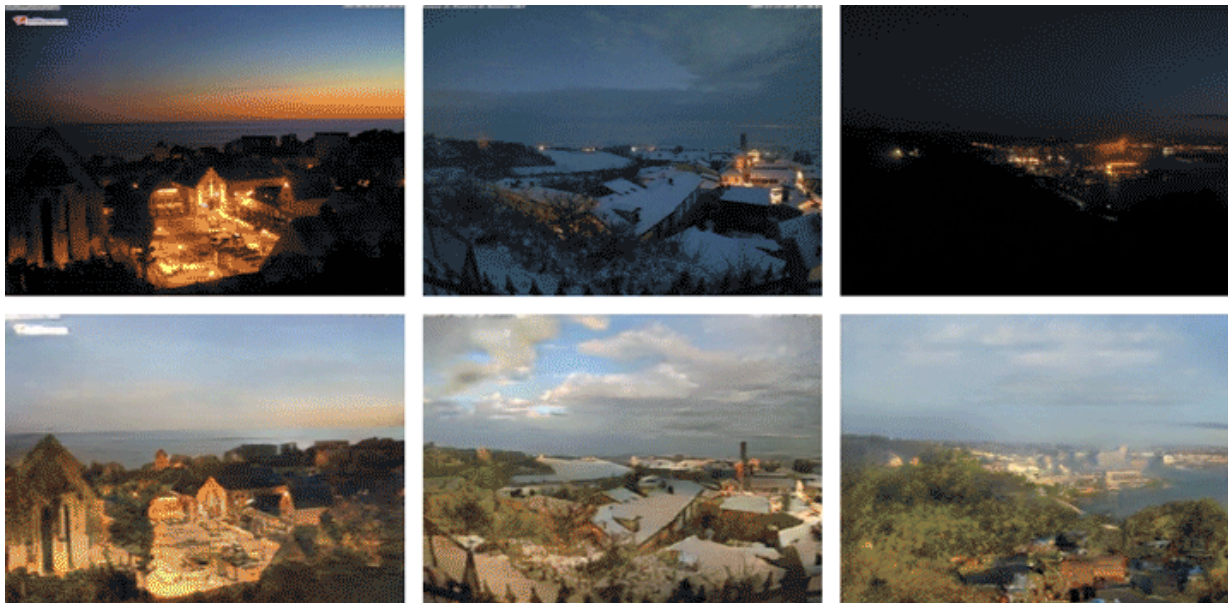
$$\begin{aligned} \mathcal{L}(G, F, D_X, D_Y) = & \mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) \\ & + \mathcal{L}_{\text{GAN}}(F, D_X, Y, X) \\ & + \lambda \mathcal{L}_{\text{cyc}}(G, F), \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{\text{cyc}}(G, F) = & \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(G(x)) - x\|_1] \\ & + \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(F(y)) - y\|_1]. \end{aligned}$$



What is **BicycleGAN**?

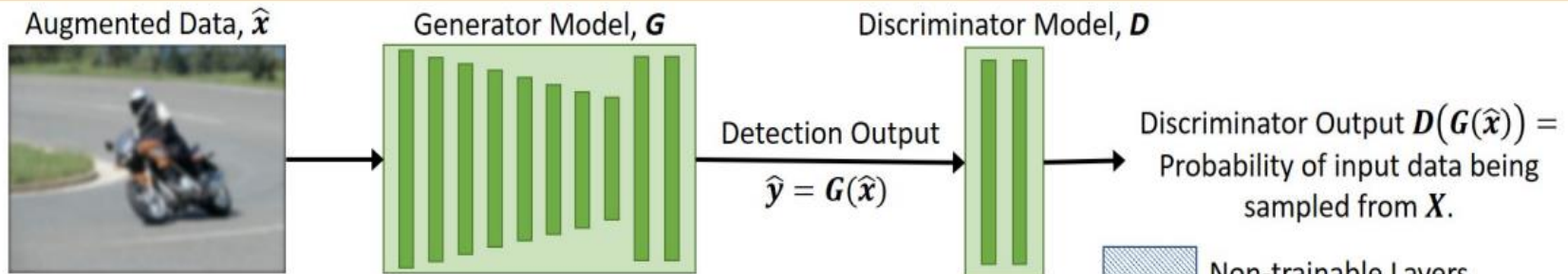
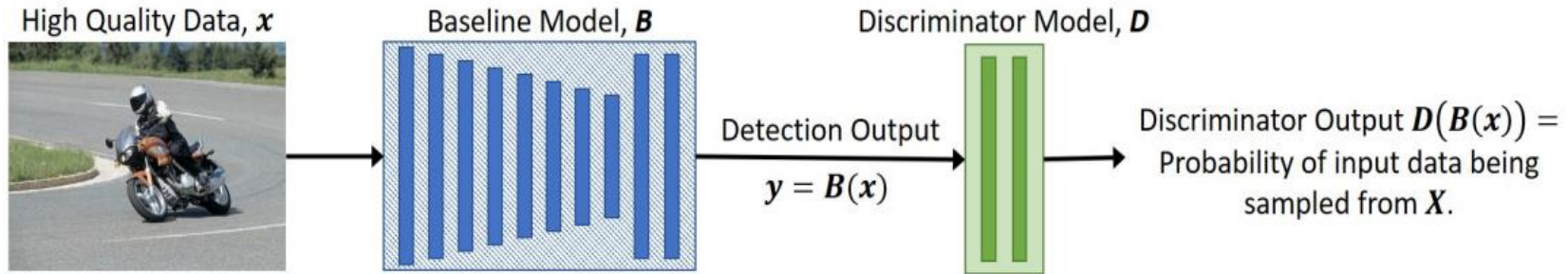
- **BicycleGAN** is a model that generates **multiple different valid outputs for the same input**.



Using GAN to improve Detection

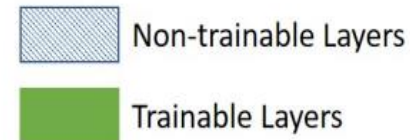
- It GAN Do Better: GAN-Based Detection of

High Quality Data: $x \in X$. Reduced Quality Data: $\tilde{x} = \rho(x) \in \tilde{X}$. Augmented Data: $\hat{x} \in \hat{X} = X \cup \tilde{X}$.



D learns to classify x and $\hat{x}$ as being sampled from X and $\hat{X}$, respectively.

G learns to outsmart D to make D classify $\hat{x}$ as being sampled from X .

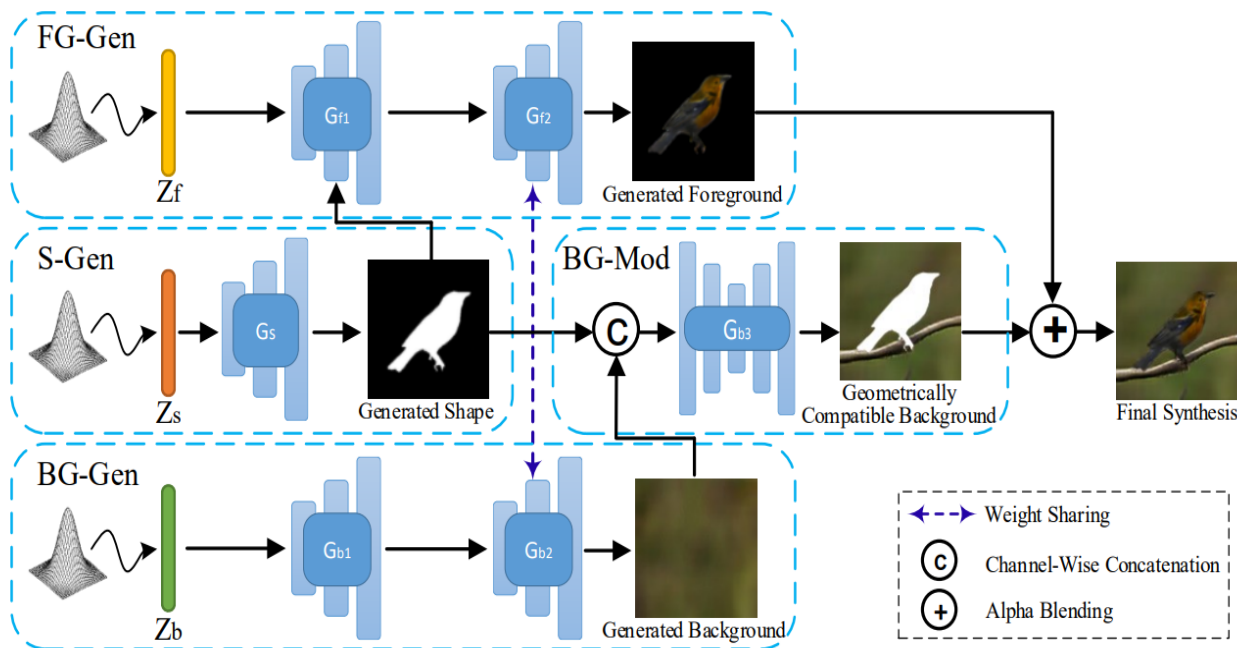


$$L_{total} = L_{OD} + \lambda L_{GAN}$$



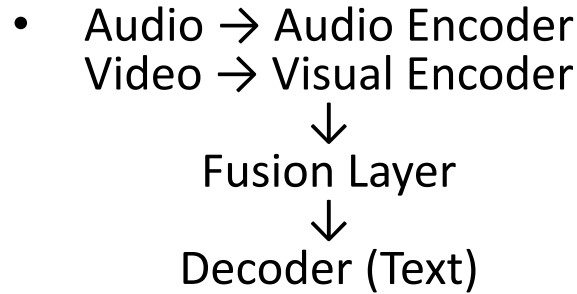
Blending GANs

- FBC-GAN: Diverse and Flexible Image Synthesis via Foreground-Background Comp**



AVSR (Audio-Visual Speech Recognition)

- Combine **audio + lip movement (video)** → generate **speech text**
- **Why AVSR?**
 - Audio alone fails when:
 - noisy environment
 - overlapping speakers
 - Video helps:
 - lip reading



Key Challenges

1. Temporal Alignment

- Audio and video must sync

2. Noise Robustness

- Audio may be corrupted

3. Feature Fusion

- How to combine modalities?

4. Data Scarcity

- Multimodal datasets are limited



CORE TECHNICAL CHALLENGES IN MULTIMODAL LEARNING

Input (Image/Text/Audio)



Representation → Alignment → Fusion →
Translation → Co-learning



Output



CORE TECHNICAL CHALLENGES IN MULTIMODAL LEARNING

1. REPRESENTATION LEARNING

- How do we convert different modalities into **meaningful numerical vectors**?

Modality	Raw Form
Image	pixels
Text	words
Audio	waveform

- Convert all modalities into a **shared embedding space**



CORE TECHNICAL CHALLENGES IN MULTIMODAL LEARNING

2. TRANSLATION (Cross-modal Generation)

Convert one modality into another

- **Examples**

- Image → Text (captioning)
- Text → Image (generation)
- Audio → Text (speech recognition)

“Translation is not exact mapping—it’s *semantic understanding*.”



CORE TECHNICAL CHALLENGES IN MULTIMODAL LEARNING

3. ALIGNMENT

Identify **which parts of one modality correspond to another**

- Sentence:
 - “A dog is playing with a ball”
- Image:
 - dog region
 - ball region
- Alignment:
 - “dog” $\leftrightarrow$ dog pixels
 - “ball” $\leftrightarrow$ ball pixels



CORE TECHNICAL CHALLENGES IN MULTIMODAL LEARNING

- 4. FUSION
 - Combine multiple modalities into a **single representation**
- 1. **Early Fusion**
 - Combine features early
- 2. **Late Fusion**
 - Combine predictions
- 3. **Hybrid Fusion**
 - Combine at multiple stages



CORE TECHNICAL CHALLENGES IN MULTIMODAL LEARNING

- 5. CO-LEARNING
 - Learning **from one modality to improve another**
- Learn image features using text captions Learn audio using video (lip reading)

1. Co-training

- train jointly using multiple modalities

- **2. Self-supervised learning**

- one modality supervises another



End of Slide

THANK YOU

